Software Requirements Specification

1. Introduction

Product Scope

The Musicmate app is a social media web application combining Spotify and Instagram features. It aims to connect users based on their music preferences.

Product Value

Musicmate provides a platform for music enthusiasts to interact, discover new music, and connect with like-minded individuals.

Intended Audience

The target users of Musicmate are primarily youth and music enthusiasts who are interested in socializing through music.

Intended Use

Users can create profiles, share music, follow others, create playlists, and engage in music-related discussions on the platform.

Definitions and Acronyms

- API: Application Programming Interface
- PHP: Hypertext Preprocessor
- MySQL: Structured Query Language
- HTML: Hypertext Markup Language
- CSS: Cascading Style Sheets
- JavaScript: Programming language for web development

2. System Requirements and Functional Requirements

System Requirements

- Cross-platform web application using HTML, CSS, JavaScript, PHP, and MySQL.
- Custom API integration with Spotify for music data.
- Scalable performance to accommodate a growing user base.
- Hybrid environment with on-premises and cloud components.

Functional Requirements

- User registration and profile creation
- Music sharing and playlist creation
- Social features like following, liking, and commenting
- Integration with Spotify API for music recommendations
- Customizable widgets for user profiles

3. External Interface Requirements

User Interfaces

- Interactive and visually appealing user interface
- Customizable widgets for user profiles
- Easy navigation and intuitive design

Hardware Interfaces

- Compatible with standard web browsers on desktop and mobile devices

Software Interfaces

- Integration with Spotify API for music data
- Backend logic for data handling and user authentication

Communication Interfaces

- Real-time messaging and notifications for user interactions

4. Non-Functional Requirements (NFRs)

Security

- Implement secure user authentication and data encryption
- Regular security audits to prevent data breaches

Capacity

- Support for handling large media files and user-generated content
- Scalable architecture to accommodate increasing data storage needs

Compatibility

- Cross-platform compatibility for web browsers and devices
- Responsive design for mobile devices

Reliability and Availability

- High availability with minimal downtime for maintenance
- Backup and recovery mechanisms for data protection

Scalability

- Ability to scale the application infrastructure based on user growth
- Performance optimization for efficient resource utilization

Maintainability

- Modular code structure for easy maintenance and updates
- Documentation for developers and administrators

Usability

- User-friendly interface with intuitive navigation
- Personalization features for a tailored user experience

By following these software requirements, the Musicmate app can be developed to meet the needs of its target audience and provide a seamless social music experience.